The Changes in Synergistic Patterns of Peri-Shoulder Muscles During Shoulder Abduction and Flexion in Patients With Rotator Cuff Tears

Miao-Qin Zhan, Jian-Ning Sun, Nan Zheng, Yu-Rong Li, and Peng Chen

***Abstract*— Understanding muscle synergy variability and its clinical relevance in rotator cuff tear (RCT) patients is crucial for elucidating motor control mechanisms and informing rehabilitation. This study uses non-negative matrix factorization (NMF) to assess the influence of age and pathological factors on synergy patterns during abduction (ABD) and flexion (FL) tasks. Fifteen young controls (YC), fifteen elderly controls (EC), and twenty elderly RCT patients were recruited. Surface electromyography (sEMG) signals from eight shoulder muscles were recorded for NMF analysis, and the correlation with the Constant-Murley Score (CMS) was evaluated via regression. Results revealed that the number of synergies in the FL task was significantly lower in the RCT group compared to the EC (4.5 ± 1.3 vs. 5.6 ± 1.1, $P = 0.01$) and YC groups (4.5 ± 1.3 vs. 5.6 ± 1.0, $P = 0.03$), with no significant difference in the ABD task ($P = 0.10$). In ABD, synergy pattern 1 was dominated by the upper trapezius and middle trapezius, while in the EC group, pattern 4 was driven by the supraspinatus, and in the YC group, both supraspinatus and posterior deltoid were involved. In FL, the middle deltoid weight was significantly increased in pattern 3 in the RCT group, suggesting compensatory activation or a pathological marker. The root mean square value of middle deltoid strongly correlated with CMS (ABD: $R^2 = 0.53$, $P < 0.001$; FL: $R^2 = 0.46$, $P < 0.001$). These findings reveal task-specific compensation mechanisms in RCT patients and provide insights for targeted rehabilitation strategies.**

***Index Terms*— Electromyography, muscle synergy, non-negative matrix factorization, rotator cuff tear.**

I. Introduction

THE shoulder joint, which has the greatest range of motion among all human joints, relies heavily on the coordinated control of the rotator cuff muscles and surrounding musculature for its functionality. Rotator cuff tear (RCT) is a common degenerative condition, with its incidence notably increasing in the aging population. Approximately 20%–30% of individuals over the age of 60 show pathological changes in the rotator cuff globally [1], [2]. These injuries not only cause pain and restrict shoulder mobility but may also lead to secondary complications, such as glenohumeral joint instability, subacromial impingement, and osteoporosis [3], [4]. Currently, clinical evaluations focus primarily on anatomical abnormalities (e.g., tendon rupture extent and muscle atrophy severity) or rely on subjective functional scales [5], [6], [7]. However, the compensatory adaptations in neuromuscular control strategies and their potential effects on upper limb motor function remain poorly understood.

Previous studies have shown that the central nervous system (CNS) simplifies multi-muscle control through a modular strategy known as muscle synergy, where a limited number of fixed activation patterns are combined to produce complex movements [8], [9]. This strategy demonstrates high adaptability in healthy individuals; however, it may undergo reorganization under pathological conditions. For instance, in patients with knee osteoarthritis (KOA), the number of synergistic modules is reduced during gait, suggesting a compensatory simplification of neuromuscular control [10]. Despite this, the adaptive mechanisms of muscle synergy following shoulder joint injury—a structure characterized by multiple degrees of freedom and high dynamic stability requirements—remain poorly understood. Most research has focused on laboratory analyses involving either healthy individuals or stroke patients [11], [12], [13]. However, the synergy patterns

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This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Committee of the First Affiliated Hospital of Fujian Medical University (MRCTA, ECFAH of FMU | 2023 | 455).

Miao-Qin Zhan, Nan Zheng, and Yu-Rong Li are with the School of Electrical Engineering and Automation and Fujian Provincial Key Laboratory of Medical Instrument and Pharmaceutical Technology, Fuzhou University, Fuzhou 350108, China (e-mail: 230127163@fzu.edu.cn; zn190120078@163.com; liyurong@fzu.edu.cn).

Jian-Ning Sun and Peng Chen are with the Department of Sports Medicine and the Department of Orthopedic Surgery, National Regional Medical Center, Binhai Campus of the First Affiliated Hospital, Fujian Medical University, Fuzhou 350005, China (e-mail: jssjn0616@163.com; 13506989645@163.com).

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exhibited by patients with rotator cuff tears during daily functional activities, and their potential association with functional prognosis, have yet to be fully explored.

Age-related declines in neuromuscular function may exacerbate compensatory mechanisms associated with rotator cuff tears. Studies have shown that healthy older adults exhibit delayed muscle activation timing and increased complexity of synergy modules during motor tasks [14], [15]. Rotator cuff tears, especially those involving the supraspinatus muscle, can lead to dysfunction and excessive compensation by the trapezius and serratus anterior muscles, resulting in abnormal synergy patterns. While these compensatory mechanisms may temporarily preserve joint function, they can also accelerate joint degeneration due to imbalanced load distribution [16], [17]. Therefore, it is crucial to differentiate the effects of aging from those of pathological conditions when analyzing changes in muscle synergy structure to develop targeted and effective rehabilitation strategies. The Constant-Murley Score (CMS) is the most commonly used subjective scale for clinical assessment of shoulder joint function. It integrates four dimensions: pain (0-15 points), daily living function (0-20 points), range of motion (0-40 points), and muscle strength (0-25 points), with a total score of 100 points. The lower the score, the more severe the functional impairment. This scale has been widely validated for its reliability and validity and can comprehensively reflect the clinical symptoms and functional status of patients with rotator cuff tears. In this article, by analyzing the correlation between CMS scores and electromyographic features, we aim to establish the association between objective neuromuscular indicators and subjective functional assessment.

In recent years, muscle synergy analysis has emerged as a valuable tool for investigating neuromuscular control mechanisms. This approach extracts synergy modules from multi-channel surface electromyography (sEMG) signals using non-negative matrix factorization (NMF), providing a quantitative assessment of CNS control strategies [18]. In neurological conditions such as stroke [19] and spinal cord injury [20], alterations in the number and composition of synergy modules have been shown to correlate closely with motor impairments. However, in musculoskeletal disorders such as rotator cuff tear, the application of muscle synergy analysis remains in its early stages. Biomechanical studies suggest that shoulder joint stability relies on the dynamic coordination between the rotator cuff muscles and the scapular stabilizers [21]. Therefore, synergy patterns may serve as a critical link between neural control mechanisms and biomechanical function.

This study aims to investigate the reorganization mechanisms of neuromuscular coordination patterns in patients with rotator cuff tears, addressing the limitations of traditional structural assessments and providing new insights into compensatory motor control pathways in degenerative shoulder joint conditions. The findings are expected to differentiate age-related physiological decline from pathology-specific synergy alterations, offering a theoretical foundation for the prevention of secondary joint injuries and the development of function-oriented rehabilitation strategies. The objectives are to clarify: (1) the independent effects of aging and pathological factors on the number of muscle synergies; (2) the abnormal spatial composition and altered temporal coordination of synergy patterns in RCT patients; (3) the relationship between the activation characteristics of rotator cuff muscles in RCT patients and CMS. We hypothesize that age and pathology exert distinct and independent influences on shoulder muscle synergy patterns. Additionally, motor function compensation in RCT patients is achieved through spatial synergy reorganization, and the activation features of compensatory muscles are significantly correlated with CMS.

TABLE I
THE PARTICIPANT DEMOGRAPHICS

Subject	RCT（n=20)	EC（n=15）	YC（n=15）
Age（yr）	64.15±6.68	62.2±4.43	19.93±0.89
Sex（M/F）	9/11	7/8	3/12
Laterality（L/R）	6/14	11/4	7/8
CMS	62.55±9.18	-	-

II. MATERIALS AND METHODS

A. Subjects

This study recruited 15 young controls (YC), 15 elderly controls (EC), and 20 patients with rotator cuff tears (RCT) who met the inclusion criteria. In this study, subjects with a BMI < 18.5 kg/m^2 (underweight) or ≥ 30 kg/m^2 (severe obesity) will be excluded. All participants voluntarily enrolled in the study and provided written informed consent. Inclusion criteria for the RCT group included persistent shoulder pain and functional impairment lasting at least six months, with no significant improvement following conservative treatment. The severity of shoulder dysfunction was assessed using the CMS. The EC group consisted of age-matched healthy individuals, while the YC group included young healthy participants without a history of shoulder injury, pain, or functional impairment. Exclusion criteria included the presence of neurological disorders, a history of shoulder surgery, or any musculoskeletal condition affecting upper limb function. To evaluate the association between EMG features and functional status, CMS were collected from patients with rotator cuff tears. The CMS assesses four domains: (1) pain, (2) activities of daily living, (3) range of motion, and (4) muscle strength. Demographic data for all participants are summarized in Table I. This study was approved by the Ethics Committee of the First Affiliated Hospital of Fujian Medical University (MRCTA, ECFAH of FMU | 2023 | 455).

B. Experimental Data

The experiment was conducted in a quiet, enclosed room to minimize external noise interference. Each participant performed two sets of task-specific movements [22], [23], [24]. A 60-second rest period was provided between sets, during which the examiner demonstrated the task and instructed participants

Fig. 1. Action schematic diagram: Left: Maximum external abduction (ABD), Right: Maximum flexion (FL).

to replicate the movements as accurately as possible in the subsequent trials. Each movement was repeated five times. All tasks commenced from a standardized starting position: the participant stood upright with both arms relaxed naturally at the sides. Each task was performed using the affected upper limb and included two movements: maximum shoulder abduction (ABD) and maximum forward flexion (FL) (Fig.1).

The FL and ABD tasks were selected because they are fundamental shoulder movements directly linked to the primary functional impairments experienced by patients with rotator cuff tears, such as dressing and raising the arm. These tasks also exhibit clear biomechanical distinctions: abduction depends on the supraspinatus-trapezius force couple, which is prone to compensation by the trapezius and serratus anterior muscles, while forward flexion demands greater stability from the subscapularis muscle and is more likely to reveal coordinated muscle reconstruction. Furthermore, these movements closely align with the range-of-motion tests used in the clinical gold standard, the CMS, and with movements explored in previous rotator cuff studies [11]. This ensures the functional relevance, pathological specificity, and comparability of the results with existing study. sEMG data were collected from eight muscles: the anterior deltoid (AD), middle deltoid (MD), posterior deltoid (PD), supraspinatus (SSP), upper trapezius (UT), middle trapezius (MT), lower trapezius (LT), and serratus anterior (SA).

C. Data Collection and Preprocessing

The data were collected using the MyoResearch ™wireless sEMG system from NORAXON (USA) with a sampling frequency of 1500 Hz. Prior to data acquisition, the skin was cleaned with 75% medical alcohol to remove dead skin cells and minimize signal interference. After the alcohol had fully evaporated, eight sEMG electrodes were placed on the participants' dry skin to record muscle activity. Before electrode placement, the target muscles were located using anatomical landmarks and confirmed through specific muscle activation tasks [25]. The electrodes were then positioned at the mid-belly of each target muscle, aligned with the direction of the muscle fibers, to ensure accurate sEMG signal acquisition.

sEMG data were processed using MATLAB R2020b. Due to environmental noise and other artifacts, raw EMG signals required preprocessing before further analysis. The processing steps were as follows: first, a fourth-order Butterworth band-pass filter with a bandwidth of 10–500 Hz was applied to remove broadband noise; second, a 50 Hz notch filter was used to eliminate power line interference. The signal was then full-wave rectified and smoothed with a low-pass filter to extract the signal envelope. Finally, min-max normalization was applied to standardize the data for subsequent analysis.

D. Detection Method for Activation Sequences

The temporal sequence of muscle activation reflects the motor control strategy of the nervous system. To identify abnormal movement patterns in patients, this study employed an adaptive two-step approach to extract activity segments, using the onset of each segment as the initial point of muscle activation. In the first step, raw EMG signals were processed with a third-order Butterworth low-pass filter with a cutoff frequency of 2 Hz to eliminate spike noise and derive the muscle activation envelope. Turning points were then identified by analyzing fluctuations in the envelope amplitude. A series of thresholds, defined as an arithmetic progression between the maximum and minimum turning points, was used to determine the number of intersections with the envelope curve. The mean threshold corresponding to the most stable intersection count was selected as the adaptive threshold, and the initial onset point was extracted to minimize the influence of false peaks. To further approximate the true onset of muscle activity, cumulative energy and energy ratio curves were computed for each activity segment. The point showing the greatest difference between these two curves was identified as the final activation onset. For detailed methodology, refer to [26]. As the duration of task execution varied across subjects, resting segment data were simultaneously extracted and used to pad the activity data, ensuring uniform action duration across participants. The standardized data were subsequently used for NMF analysis.

E. Extraction of Muscle Synergistic Actions

The schematic diagram of muscle synergy is shown in Fig. 2 In muscle synergy analysis, the non-negative matrix V is factorized into two non-negative matrices, W and H, using a multiplicative update algorithm. In electromyographic studies, the preprocessed high-dimensional surface EMG data are typically represented as matrix V, where all elements are non-negative. The goal of NMF is to identify a small set of non-negative muscle synergy patterns. The decomposition results in two matrices: the synergy matrix W and the activation coefficient matrix H, which together characterize the spatiotemporal features of muscle synergies. The synergy matrix W represents the relative contributions of individual muscles to each synergy module. Each element in W indicates the extent to which a specific muscle contributes to a given synergy, suggesting that a single muscle may participate in multiple synergies [27]. In contrast, the activation coefficient matrix H describes the temporal evolution of synergy patterns.

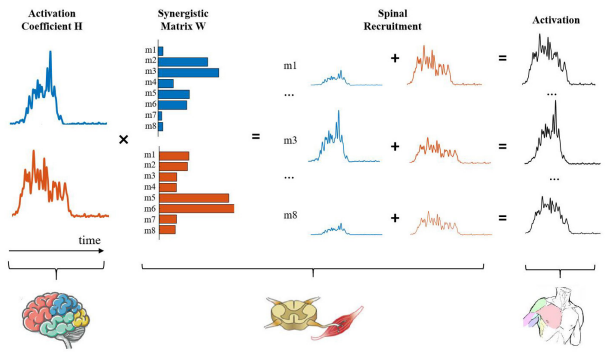

Fig. 2. Schematic Diagram of Muscle Synergistic Action. Different colors represent different synergy modes. On the left side, the blue and red curves represent the activation coefficients of different synergy patterns, reflecting the dynamic regulation process of the nervous system on the corresponding muscle synergy patterns over time. The middle column chart shows the weight distribution of each synergy pattern on different muscle channels, where m1-m8 represent the different muscle channels collected. The middle blue and red curves are the "recruitment curves", obtained by multiplying the weight of each synergistic muscle with its corresponding activation coefficient, showing the relative contribution of each muscle after a specific synergy is activated. The right side of the figure is the actual muscle activation signal, which is the linear superposition result of the activation effects of all synergy patterns.

Each element in H corresponds to the activation level of a particular synergy at a specific time point, reflecting the temporal dynamics of muscle coordination. NMF can be expressed as follows:

$$V_{ij} \approx (WH)_{ij} = \sum_{\mu=1}^{n} W_{i\mu} H_{\mu j} = V'_{ij} \tag{1}$$

where V (m × t matrix; where m is the number of muscles and t is the number of time points) is a linear combination of the synergy matrix W (m × n matrix; where n is the number of modules) and the weighted coefficient matrix H (n × t matrix), and V' represents the reconstructed surface electromyography signal.

F. Determination of the Number of Synergy

Determining the appropriate number of muscle synergies is a critical step in synergy analysis. An insufficient number of synergies may fail to capture the full scope of motor control information, while an excessive number may introduce redundancy and reduce interpretablity [11]. A commonly used criterion for selecting the optimal number of synergies is the Variance Accounted For (VAF). The definition of VAF is as follows:

$$VAF = 1 - \frac{\sum (V - V')^2}{\sum (V)^2} \tag{2}$$

where V represents the electromyographic signal, and V' represents the reconstructed electromyographic signal.

In this study, the number of synergy effects gradually increased from 1 to 8. When the VAF exceeded 90% and the gain of adding one more synergy VAF was less than 2%, it was determined as the optimal number [28], [29]. This standard was applied to the electromyographic data of each subject for each action, ensuring the robustness between the actions. For each synergy number, the experiment was repeated 20 times.

G. Defined Muscle Module Group

Based on previous studies [30], [31], muscle synergy modules were sorted according to the peak timing of their corresponding temporal activation profiles.

H. Correlation Analysis

To investigate the relationship between EMG features and the CMS in patients with rotator cuff tears, a linear regression model was used to examine their linear association. The results were visualized through scatter plots and fitted regression curves. A 95% confidence interval was established, and the goodness of fit was assessed using the coefficient of determination (R^2). The statistical significance of the model was evaluated using the F-test and corresponding p-value, with significance set at $p < 0.05$.

I. I. Statistical Analysis

In this study, the Shapiro–Wilk test and Levene's test were used to assess whether the coordination measures of each group (RCT, EC and YC) met the assumptions of normality and homogeneity of variance. If both assumptions were satisfied across all three groups, a one-way ANOVA was applied to evaluate differences in the number of muscle synergies among the groups. If normality was satisfied but variance homogeneity was violated, Welch's ANOVA was used instead. When normality was not met, the Kruskal–Wallis H test was applied. When significant differences were detected, post hoc multiple comparisons were performed to identify specific group differences. Statistical significance was indicated by p <0.05.

III. Results

A. Muscle Synergy Number

The schematic diagram showing the extraction of the cooperative quantity of one of the subjects is shown in Fig. 3. The results of the synergy number analysis are shown in Fig. 4. In the NMF analysis, during the execution of the ABD task, the RCT group extracted 4.3 ± 1.7 muscle modules, the EC group 4.1 ± 1.2, and the YC group 5.5 ± 1.2. No statistically significant difference was observed in the number of modules among the three groups ($p = 0.10$). However, during the forward FL task, the RCT group extracted significantly fewer modules (4.5 ± 1.3) compared to both the EC group (5.6 ± 1.1, $p = 0.01$) and the YC group (5.6 ± 1.0, p = 0.03). No significant difference was found between the EC and YC groups (p= 0.80).

B. Muscle Synergistic Pattern

Fig. 5 and Fig.6 show the temporal activation sequences and muscle weightings during the ABD and forward FL tasks for each group. The number of synergy modules was fixed to the average number for each group, and synergy patterns were labeled according to the order of temporal activation.

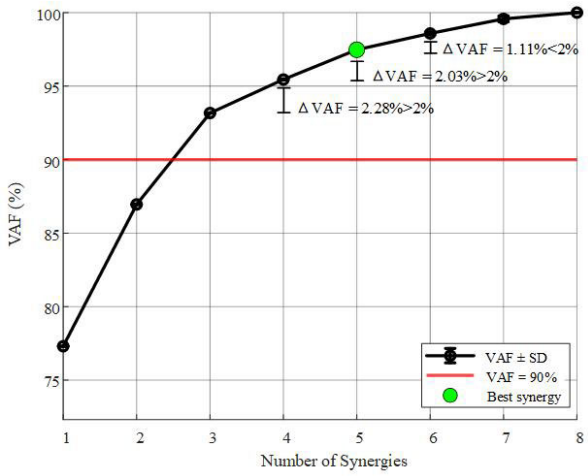

Fig. 3. A schematic diagram showing the number of concurrent actions extracted by one of the subjects. For each synergy number, the experiment was repeated 20 times.

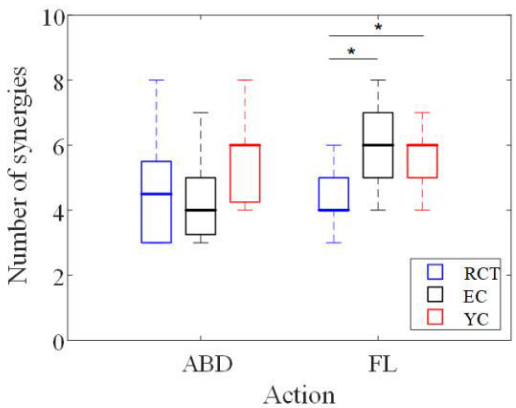

Fig. 4. The number of synergy actions performed by each group when executing different actions (blue: RCT group, black: EC group, red: YC group). The bold horizontal lines indicate the median of each data group, the boxes represent the quartiles of each data group, and the dotted lines above and below show the maximum and minimum values respectively.

Fig. 7 display the normalized muscle weight coefficients for the mean synergy patterns during the ABD and FL tasks, with normalization performed relative to the maximum value within each group.

During the execution of the ABD task, the average number of muscle synergy modules was five in both the RCT and EC groups, and six in the YC group. In Mode 1, the UT exhibited the highest weight in the RCT group, followed by the MT, AD, and MD. In contrast, the EC group showed the highest contribution from the SSP, while the YC group was primarily dominated by the SSP and PD. In Mode 2, the RCT group showed a decrease in the contribution of the UT and an increase in the SSP and AD, which became dominant. The EC group exhibited increased weighting of the UT and SA, while the YC group remained dominated by the SSP. In Mode 3 and Mode 4, the SA was dominant in the RCT group. In Mode 3, the EC and YC groups showed distinct muscle dominance: the EC group was dominated by the SA and UT, while the YC group showed higher weights for the PD, MT, and SSP. In Mode 4, both the EC and YC groups displayed similar dominant muscle contributions, primarily involving the AD, MD, and both the UT and MT. In Mode 5 for the RCT and EC groups, and in Mode 5 and Mode 6 for the YC group, all muscles exhibited approximately balanced weight distributions, indicating a more uniform contribution across muscle groups.

During the execution of the forward FL task, the average number of muscle synergy modules was five in the RCT group and six in both the EC and YC groups. In Mode 1, the RCT group showed high contributions from the SSP, UT and PD. In the EC group, the dominant contributors were the UT and SSP, while in the YC group, the PD was most prominent. In Mode 2, the RCT group exhibited a decrease in the contributions of the UT and SSP, with an increased contribution from the SA, which became dominant. Similarly, in the EC group, both the UT and SA remained dominant. In contrast, the YC group showed a more diverse pattern, with substantial contributions from the PD, UT, and MT, SA and SSP. In Mode 3, the MD muscle had the highest weight proportion in the RCT group, while the UT, MT, and the AD had relatively high proportions.

In the EC and YC groups, the SA and SSP were the primary contributors, respectively. In the RCT group, Mode 4 and Mode 5 were dominated by the SA and UT. In Mode 4 of the EC group, the dominant muscles were the MT and LT muscles, while in the YC group, the dominant muscles were the AD and MD. In Mode 5 and Mode 6 of the EC group and the YC group, all muscles showed approximately balanced weight proportions.

C. The Correlation Between the Muscle Synergy Characteristics of the RCT Group and the CMS

The relevant data are presented in Fig. 8 - 10. During the evaluation of shoulder joint function, a significant correlation was observed between the EMG activity of each division of the deltoid muscle and the CMS in the RCT group. Specifically, during the ABD task, the root mean square (RMS) values of the AD, MD, and PD were linearly correlated with the CMS ($R^2 = 0.37$, 0.53, and 0.27, respectively; all $p < 0.05$). Similar significant correlations were found during the forward FL task ($R^2 = 0.34$, 0.46, and 0.34; all $p < 0.05$). Notably, the MD exhibited the strongest correlation in both tasks (ABD: $R^2 = 0.53$; FL: $R^2 = 0.46$; both $p < 0.001$). In contrast, the number of muscle synergy patterns showed no significant correlation with the CMS ($R^2 = 0.00$, $p = 0.91$ for ABD; $R^2 = 0.01$, $p = 0.69$ for FL). Additionally, no significant correlations were found between the RMS values of the trapezius, serratus anterior, or supraspinatus muscles and the CMS (all $p > 0.05$).

IV. Discussion

A. Summary

This study compared muscle synergy patterns during shoulder FL and ABD tasks across three groups: YC, EC and RCT. The analysis revealed distinct effects of aging and pathology

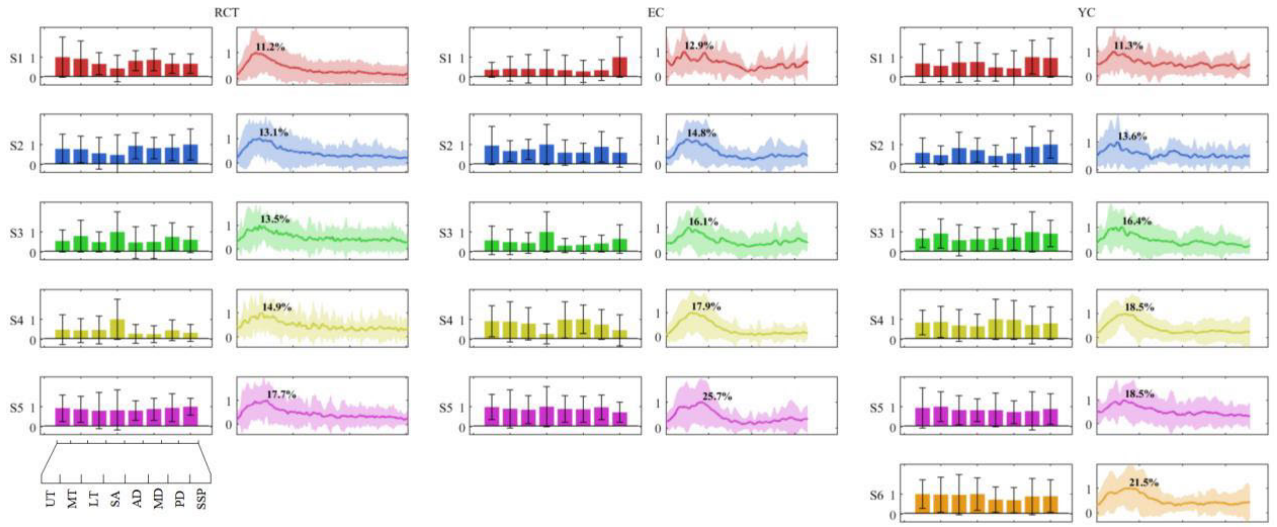

Fig. 5. Weight and temporal sequence of muscle activation during the ABD movement task, (left: RCT group, middle: EC group, right: YC group), naming the synergy patterns according to the sequence of activation, and representing the same sequence of synergy patterns in each group with the same color. The left side of each group's figure represents the muscle weights of each synergy pattern, the height of the columns represents the average value of each group's data, the error bars represent the standard deviation of each group's data, and the right side's figure represents the temporal sequence of activation of each synergy pattern, the solid lines represent the average values of each group's data, and the shaded parts represent the standard deviation of each group's data.

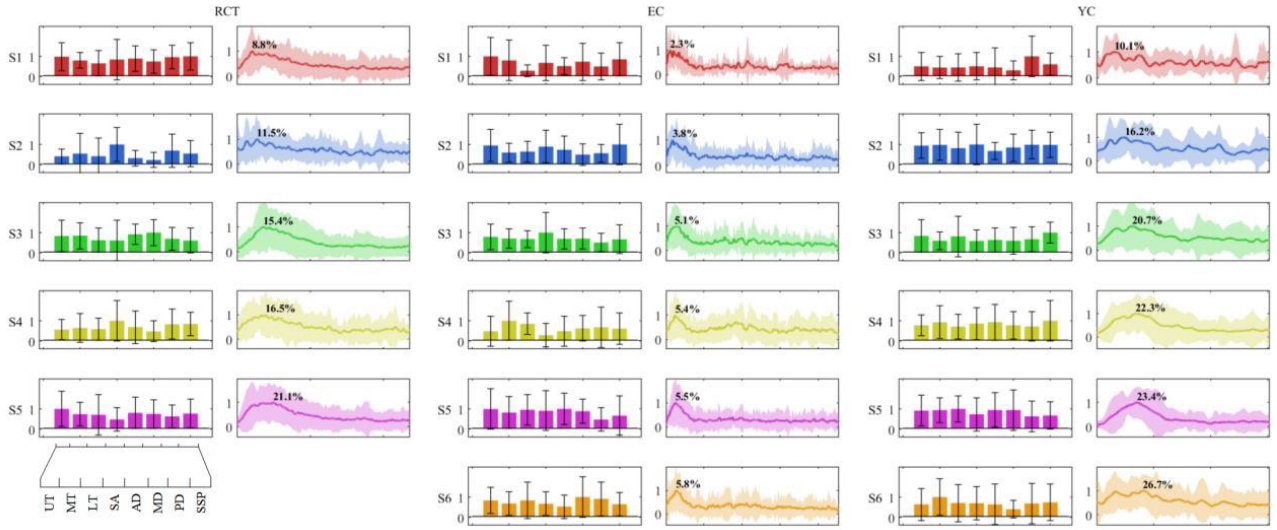

Fig. 6. Weight and temporal sequence of muscle activation during FL task, (left: RCT group, middle: EC group, right: YC group), naming the synergy patterns according to the sequence of activation, and representing each synergy pattern of each group with the same color. The left side of each group's figure represents the muscle weights of each synergy pattern, the height of the column indicates the average value of each group's data, the error bar indicates the standard deviation of each group's data, and the right side's figure represents the temporal sequence of activation of each synergy pattern, the solid line indicates the average value of each group's data, and the shaded part indicates the standard deviation of each group's data.

on neuromuscular control strategies. The number of synergy modules during the FL task was significantly lower in the RCT group compared to both the EC ($p = 0.01$) and YC ($p = 0.03$) groups, whereas no significant difference was observed among the groups during the ABD task. These results suggest that the reduction in synergy number associated with rotator cuff tear is task-specific and that aging alone does not significantly affect synergy quantity. Further analysis of synergy composition demonstrated that both age and pathology influence muscle activation patterns. In the ABD task, the RCT group exhibited characteristic spatial reorganization, with the UT and MT showing the highest weights in Mode 1. In contrast, the EC group was dominated by the SSP, while the YC group was characterized by high contributions from both the SSP and PD. In Mode 4, the RCT group was predominantly characterized by the AD, whereas the EC and YC groups showed greater involvement of the AD and MD as well as the UT and MT. During the FL task, the MD demonstrated significantly increased weighting in Mode 3 in the RCT group, suggesting that compensatory deltoid

activation may serve as a pathological marker. Moreover, the RMS values of EMG signals from the three deltoid

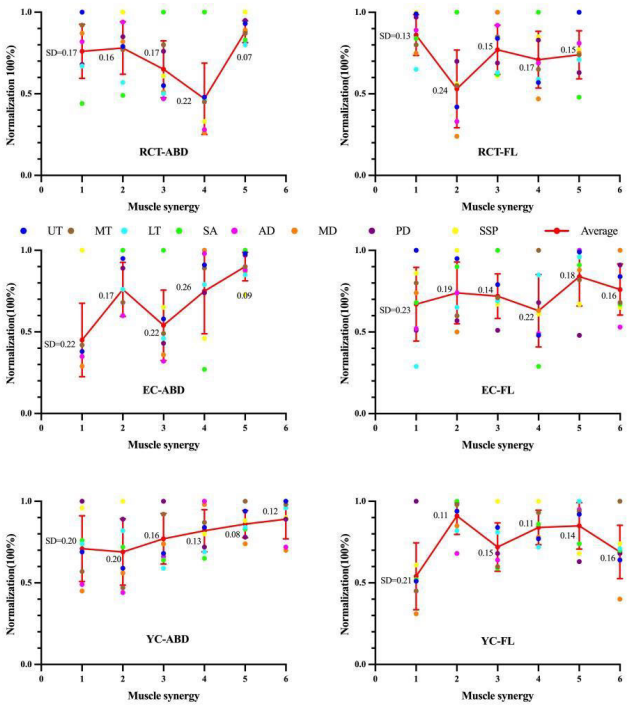

Fig. 7. Muscle weight coefficients under different cooperative modes in each group (the first row: RCT group, the second row: EC group, the third row: YC group). Each point represents the value of each muscle, and SD represents the standard deviation of the weight coefficients of all muscles. The smaller the SD, the more average the contribution of all muscles.

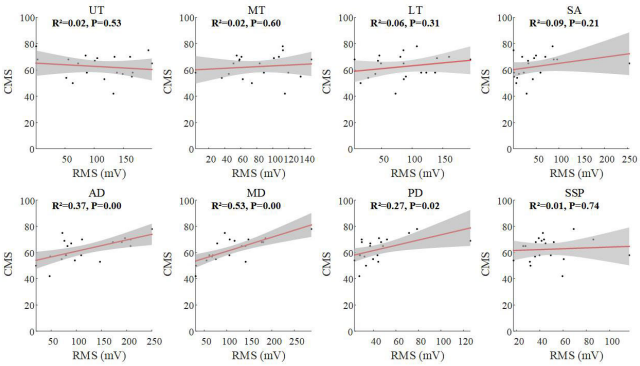

Fig. 8. Linear regression of the RMS values of the muscles when performing ABD movements and the CMS was conducted, with each point representing the value for each patient. The solid line represents the fitting curve, the shaded part represents the confidence interval, R2 represents the determination coefficient in the linear regression model, and P represents the statistical significance of the linear regression model.

branches showed significant linear correlations with the CMS. In contrast, the number of synergy modules was not significantly correlated with CMS in either task (ABD: $R^2 = 0.00$; FL: $R^2 = 0.01$). These findings support the hypothesis that functional compensation in rotator cuff injury is achieved through spatial reorganization of muscle coordination, rather than merely by reducing the number of synergies. This study, for the first time, reveals the task-dependent nature of coordination control mechanisms in musculoskeletal disorders.

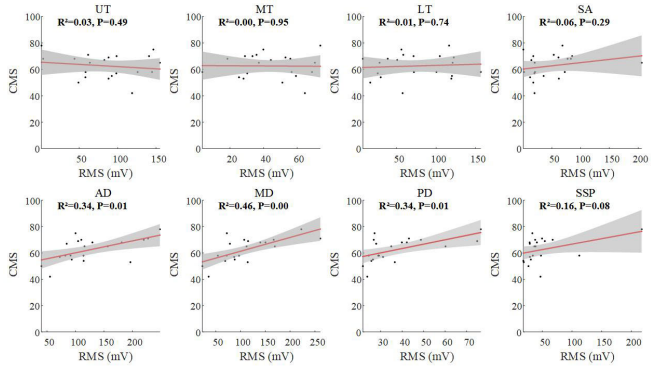

Fig. 9. Linear regression of the RMS values of the muscles when performing FL movements and CMS was conducted, with each point representing the value for each patient. The solid line represents the fitting curve, the shaded part represents the confidence interval, R2 represents the determination coefficient in the linear regression model, and P represents the statistical siqnificance of the linear reqression model.

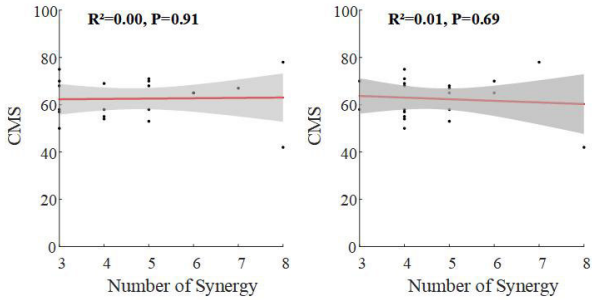

Fig. 10. Linear regression of muscle synergy quantity and CMS (left: ABD, right: FL), with each point representing the value for each patient. The solid line represents the fitting curve, the shaded part represents the confidence interval, R2 represents the determination coefficient in the linear regression model, and P represents the statistical significance of the linear regression model.

B. Task-Specific Differences in Muscle Synergy Patterns: Differential Impacts of Age and Pathological Factors

The results of this study indicate that the influence of age on muscle synergy patterns in individuals with RCT is task-specific. During the shoulder ABD task, no significant difference was observed in the number of synergy modules between the EC group and the YC group. However, during the forward FL task, the number of synergy modules in the RCT group was significantly lower than in both the EC and YC groups, with no statistical difference between the latter two. These findings suggest that age-related neuromuscular adaptations may not substantially alter the complexity of shoulder synergy control, whereas the reduction in synergy number observed in RCT patients is more likely attributable to pathology-induced compensatory mechanisms.

C. Composition of Rotator Cuff Muscles and Mechanism of Motion Compensation

The rotator cuff is composed of the supraspinatus, infraspinatus, subscapularis, and teres minor muscles. The infraspinatus and subscapularis form a force couple with the trapezius, primarily contributing to the internal and

external rotation of the shoulder joint. Simultaneously, the supraspinatus and deltoid muscles constitute another force couple responsible for shoulder elevation and depression [32]. As the primary muscle responsible for lifting the shoulder, the deltoid can elevate the rotational center of the glenohumeral joint during shoulder elevation, potentially narrowing the subacromial space and restricting the range of motion (ROM) during abduction [33], [34]. Recent studies suggest that the supraspinatus, often referred to as the "initiating muscle" of shoulder elevation, primarily functions to depress the humeral head at the start of the elevation phase, maintaining the balance of the superior-inferior force couple [16], [35]. When the supraspinatus is injured or its ability to depress the humeral head is compromised, it results in characteristic changes

observed in RCT patients and limits shoulder ROM [36]. However, research indicates that the degree of ROM limitation is not directly proportional to the extent of rotator cuff damage. This discrepancy is largely attributed to compensatory activation by surrounding muscles. Several studies [37], [38] have shown that the trapezius and pectoralis major muscles play key roles in compensatory mechanisms, particularly during the early phase of shoulder movement. Our findings support this view. In the Mode 1 during the ABD task, the healthy groups (EC/YC) demonstrated dominant activation of the SSP and PD (Fig. 4), whereas the RCT group showed elevated contributions from the UT and MT. This suggests that dysfunction of the supraspinatus leads to overactivation of the trapezius and serratus anterior muscles to initiate upward arm movement and stabilize the scapula.

D. Characteristics of Muscle Synergistic Spatial Reconstruction in Rotator Cuff Injuries

In both the ABD and forward FL tasks, the number of muscle synergies in the RCT group was consistently five, while the YC group exhibited six synergies for both tasks. The EC group demonstrated five synergies during ABD and six during FL. These results align with previous studies, which reported three to five muscle synergies during multi-directional extension [39], [40] and up to six global synergies during multi-plane shoulder movements [11]. The number of extracted muscle synergies is influenced by various factors, including biomechanical and task-related constraints [41], the number and selection of muscles analyzed [42], and the criteria used to determine the number of synergies, such as variance VAF, coefficient of determination (R^2), and z-score thresholds [43], [44].

We believe that the single-plane task cycle of the shoulder joint exhibits temporal and spatial components analogous to those of the gait cycle. Previous studies [1] have identified four muscle synergy patterns during lower-limb walking, each associated with a specific functional sub-phase of the gait cycle: forward propulsion and leg swing at the beginning of early stance (∼15% gait cycle), late stance (∼45% gait cycle), early swing (∼70% gait cycle), and late swing (∼85% gait cycle). Consistent with this stage-based framework, shoulder movements involve multiple kinematic and biomechanical stages, each imposing distinct demands on joint torque generation, scapulothoracic stability, and overall joint control. Throughout these stages, the relative contributions of primary agonist muscles, stabilizers, and antagonists shift dynamically. We believe that the biomechanical movement staging method can also be applied to the shoulder joint, with synergy patterns 1-5 or 1-6 corresponding to the ascending and descending stages of shoulder joint movement. Taking the abduction (ABD) task as an example, the synergy patterns of the scapula and humeral head generally conform to the characteristics of shoulder-humeral rhythm and can be divided into three stages. Stage 1: The initiation stage of shoulder joint lifting, mainly relying on the activation of the supraspinatus muscle to promote the rotation of the humeral head within the glenoid cavity, with only a small amplitude of scapular upward rotation. This stage aims to initiate the movement and preliminarily adjust the joint position. In the synergy mode 1, both the EC group and the YC group showed a high level of activation of the supraspinatus muscle. However, the RCT group, due to the tear of the supraspinatus muscle, the upper trapezius muscle and the deltoid muscle compensated for the initiation stage of the shoulder joint abduction movement. Stage 2: In this stage, the synergy between the scapula and humeral head increases, and the scapular rotation ratio gradually approaches 2:1. As the angle increases, the scapula begins to actively rotate upward. The activation of the supraspinatus muscle decreases, while the activation of the deltoid muscle increases to accelerate the concentric movement of the shoulder joint. The activation of the lower part of the trapezius and serratus anterior muscles increases to drive scapular upward rotation, with the upper trapezius assisting in stability. Compared to synergy pattern 1, in synergy pattern 2, both the EC and YC groups showed similar activation patterns, while the RCT group showed an increase in supraspinatus and anterior deltoid activation to maintain continued upward movement of the shoulder joint, with reduced activation of the trapezius muscle as compensation. Stage 3: The amplitude of scapular upward rotation further increases. As the shoulder joint approaches its maximum range of motion, the scapula completes its final upward rotation, while the humeral head movement decreases, ensuring that the shoulder joint reaches its maximum range of motion while maintaining stability. During this stage, the middle deltoid muscle on the humerus continues to exert force, while the infraspinatus and teres minor muscles assist in stabilizing; the upper and lower parts of the trapezius muscle on the lateral side and the serratus anterior muscle are activated to a greater extent, driving the scapula to perform a significant upward rotation and posterior tilt, achieving the maximum range of motion. It should be noted that the range of motion of the shoulder joint in patients with rotator cuff tears may not always reach 180 degrees due to disease-related limitations. In our experimental design, patients were asked to lift their arms to the maximum extent possible, so not all participants in the RCT group were able to complete the third stage of shoulder joint movement. In synergy patterns 3-4 for the YC and EC groups, peak activation of the deltoid muscle indicates the final stage of shoulder joint upward movement. The activation of the trapezius and the anterior serratus muscles increases to assist with scapular upward rotation and stability. In synergy

patterns 3-4 of the RCT group, high activation of the anterior serratus muscle was observed, which, in coordination with the upper trapezius muscle, compensates for the trapezius muscle and jointly completes the upward rotation of the scapula, providing stable support for the glenohumeral head sliding beneath the acromion and increasing the stability of the scapula. Previous studies [25], [37] have also corroborated this point. In the final stage of the abduction movement in patients with rotator cuff injuries, electromyographic activity of the anterior serratus muscle increases significantly as the angle increases, especially peaking in the 90° - 180° range. This indicates that this stage is the high activation period of the anterior serratus muscle, necessary to counteract the gravity of the scapula and maintain its attachment to the chest wall. In the descending phase of the ABD task, the shoulder joint primarily maintains eccentric muscle contractions, with the biomechanical focus on stabilizing the movement. From maximum abduction, the deltoid and supraspinatus muscles eccentrically contract to control the descent speed, while the trapezius and serratus anterior muscles eccentrically coordinate to slowly rotate the scapula downward along with the humerus. In synergy mode 5 of the RCT group, the activation of the deltoid, trapezius, and supraspinatus muscles increases to maintain stability during the shoulder joint's descending phase. In synergy modes 5 of the EC group and 5-6 of the YC group, the activation of the deltoid and supraspinatus muscles decreases compared to the previous phase, indicating the end of the shoulder joint lifting movement and the beginning of the descending phase. The relative increase in the activation of the anterior serratus muscle suggests that the shoulder joint is returning from its highest point, with the anterior serratus muscle maintaining scapular rotation. In synergy mode 6 of the YC group, the activation of all muscle groups is higher than in synergy mode 5. We believe this may be related to the continuity of the abduction task, specifically at the end of the descending phase, in preparation for the next shoulder joint lift. Further evidence is needed to verify this, such as by referring to studies like those by [11], which restrict the range of motion of the upper limb's shoulder and set different starting and ending positions. This approach would aid in analyzing muscle coordination performance in different stages of single-plane movement. However, it must be noted that interrupting the subject's action cycle may affect the final conclusions, as discussed in [46].

E. The Clinical Biomarker Value of Deltoid Muscle Activation

Although the number of muscle synergies was not significantly correlated with the CMS (ABD: $R^2 = 0.00$; FL: $R^2 = 0.01$), the RMS values of EMG activity in each branch of the deltoid muscle within the RCT group showed strong linear associations with CMS (e.g., middle deltoid [MD] during ABD: $R^2 = 0.53$, $p < 0.001$). This finding suggests that the activation intensity of the deltoid, particularly the middle fibers, may serve as an objective biomarker for evaluating shoulder joint function. Two potential mechanisms may explain this correlation: (1) compensatory overactivation of the deltoid muscle may directly reflect the degree of rotator cuff dysfunction; (2) the "muscle strength" component of the CMS is closely related to the contractile efficiency of the deltoid. Notably, RMS values from the trapezius and supraspinatus muscles were not significantly associated with CMS scores, implying that isolated strengthening of these muscles in conventional rehabilitation programs may have limited effectiveness in improving functional outcomes. These results underscore the importance of optimizing neural control strategies by integrating muscle synergy patterns into rehabilitation design.

F. Differences in Multi-Degree-of-Freedom Motion Compensation for Rotator Cuff Injuries

This study is the first to report task-dependent changes in the number of muscle synergies in patients with rotator cuff tears, specifically showing a reduction during forward FL while remaining unchanged during ABD. This finding contrasts with the general reduction in synergy complexity observed in the gait patterns of patients with knee osteoarthritis (KOA). The discrepancy may arise from the multi-degree-of-freedom nature of the shoulder joint and its diverse compensatory mechanisms. During external rotation, patients with rotator cuff injuries compensate for lost external rotation strength by increasing the activation of the middle fibers of the trapezius, the anterior serratus, and the levator scapulae muscles. Additionally, they enhance activation of the upper trapezius to substitute for the initial stabilizing role of the supraspinatus, forming a multi-level compensatory chain involving "prime movers – stabilizers – scapular controllers," thereby maintaining a near-normal synergy count. In the FL task, however, the force production of the anterior deltoid heavily relies on the stabilizing function of the subscapularis. Injury to the subscapularis disrupts this "force – stability" coupling, and due to the lack of an effective compensatory muscle group—given the limited compensatory capacity of the trapezius and anterior serratus—there is a resulting reduction in the number of synergistic modules. In patients with rotator cuff tears, this reduction likely reflects a simplified control strategy adopted by the central nervous system to preserve joint function under impaired conditions. While such compensation may temporarily stabilize the joint, it ultimately reduces the dynamic adaptability of the system due to decreased synergy redundancy [47], [48]. Notably, because the subscapularis is located deep within the body, its functional alterations are difficult to detect with conventional surface electromyography. Future studies should therefore explore the dynamic adaptive behavior of the shoulder during forward flexion and consider using motion capture and dynamic biomechanical analysis to indirectly infer changes in motor control strategies.

G. Advantages Over Traditional Assessment Methods

Although traditional imaging techniques and CMS scales can assess the structural damage and subjective function of rotator cuff tears, they cannot capture the neural-muscular compensation mechanisms [49]. In this study, through muscle synergy analysis, it was found that patients with rotator cuff tears exhibited characteristic spatial reconfiguration and

task-specific synergy changes in the activation patterns of the shoulder muscles. This finding explains the functional compensation pathways of degenerative rotator cuff tears from the neural mechanism perspective and confirms that the electromyographic activation intensity of the middle deltoid muscle can be used as an objective biomarker for assessing the functional status, providing a theoretical basis for breaking through the limitations of traditional structural assessment and optimizing neuro-muscular-guided rehabilitation strategies.

H. Research Limitations and Future Directions

This study has several limitations: (1) Although we compared muscle synergy patterns among the three subject groups, RCT, EC and YC, during shoulder flexion and abduction, and analyzed them based on activation sequences and muscle weightings, we were unable to isolate the specific contributions of aging versus pathological factors to the observed synergy alterations; (2) Our analysis primarily focused on the relationship between muscle synergy patterns and movement phases during abduction in the RCT group, and we proposed a hypothesis linking synergy activation sequences to biomechanical movement stages of the shoulder joint. However, this hypothesis remains speculative and requires further empirical validation. No stage-specific synergy patterns were observed during forward FL and thus were not discussed; (3) sEMG cannot effectively detect the activity of deep rotator cuff muscles, such as the subscapularis, which may lead to an underestimation of the true complexity of muscle synergy organization. Future research should consider integrating ultrasound imaging or high-density sEMG to investigate the involvement of deep muscle groups in synergy reorganization and to longitudinally examine how changes in synergy patterns relate to the progression of joint degeneration.

V. Conclusion

This study found that age had no significant effect on the number of shoulder joint muscle synergies. In the RCT group, the number of synergies varied in a task-dependent manner, with simplification observed only during the forward FL task. The synergy activation patterns exhibited distinct spatial reorganization, such as early activation of the trapezius at the onset of abduction. The RMS values of the MD showed strong correlations with the CMS, indicating their potential as objective biomarkers for shoulder function. These findings highlight the task-specific nature of compensatory mechanisms in rotator cuff injury and elucidate the reconstruction of a "dynamic-stabilization synergy chain," thus overcoming the limitations of traditional anatomical assessment and providing a theoretical foundation for neuromuscular control-based precision rehabilitation and the prevention of secondary injuries.

References

[1] H. Hinsley, C. Ganderton, N. K. Arden, and A. J. Carr, "Prevalence of rotator cuff tendon tears and symptoms in a chingford general population cohort, and the resultant impact on U.K. health services: A cross-sectional observational study," *BMJ Open*, vol. 12, no. 9, Sep. 2022, Art. no. e059175.

[2] T. Teunis, B. Lubberts, B. T. Reilly, and D. Ring, "A systematic review and pooled analysis of the prevalence of rotator cuff disease with increasing age," *J. Shoulder Elbow Surg.*, vol. 23, no. 12, pp. 1913–1921, Dec. 2014.

[3] I. Kawashima et al., "Prevalence and treatment rates of osteoporosis among individuals with rotator cuff tears," *J. Shoulder Elbow Surg.*, vol. 33, no. 11, pp. e606–e609, Nov. 2024.

[4] K. Takagishi et al., "Risk factors for shoulder osteoarthritis with rotator cuff tear in the elderly general population," *J. Shoulder Elbow Surg.*, vol. 31, no. 12, pp. 2562–2569, Dec. 2022.

[5] Y. Harada et al., "Prevalence of rotator cuff tears among older tennis players and its impact on clinical findings and shoulder function," *J. Sport Rehabil.*, vol. 31, no. 7, pp. 849–855, Sep. 2022.

[6] S. K. Mehta, S. A. Teefey, W. Middleton, K. Steger-May, J. A. Sefko, and J. D. Keener, "Prevalence and risk factors for development of subscapularis and biceps pathology in shoulders with degenerative rotator cuff disease: A prospective cohort evaluation," *J. Shoulder Elbow Surgery*, vol. 29, no. 3, pp. 451–458, Mar. 2020.

[7] S. I. Macías-Hernández, J. R. García-Morales, C. Hernández-Díaz, I. Tapia-Ferrusco, O. B. Velez-Gutiérrez, and T. I. Nava-Bringas, "Tolerance and effectiveness of eccentric vs. concentric muscle strengthening in rotator cuff partial tears and moderate to severe shoulder pain. A randomized pilot study," *J. Clin. Orthopaedics Trauma*, vol. 14, pp. 106–112, Mar. 2021.

[8] V. Lanzani, C. Brambilla, and A. Scano, "Kinematic-muscular synergies describe human locomotion with a set of functional synergies," *Biomimetics*, vol. 9, no. 10, p. 619, Oct. 2024.

[9] K. Zhao, Z. Zhang, H. Wen, Z. Wang, and J. Wu, "Modular organization of muscle synergies to achieve movement behaviors," *J. Healthcare Eng.*, vol. 2019, pp. 1–9, Nov. 2019.

[10] S. Liu et al., "Factors, characteristics and influences of the changes of muscle activation patterns for patients with knee osteoarthritis: A review," *J. Orthopaedic Surg. Res.*, vol. 20, no. 1, Jan. 2025, Art. no. 112.

[11] J. Umehara, M. Yagi, T. Hirono, Y. Ueda, and N. Ichihashi, "Quantification of muscle coordination underlying basic shoulder movements using muscle synergy extraction," *J. Biomechanics*, vol. 120, May 2021, Art. no. 110358.

[12] B. Pan, Z. Huang, T. Jin, J. Wu, Z. Zhang, and Y. Shen, "Motor function assessment of upper limb in stroke patients," *J. Healthcare Eng.*, vol. 2021, pp. 1–11, Feb. 2021.

[13] J.-H. Park, J.-H. Shin, H. Lee, J. Roh, and H.-S. Park, "Alterations in intermuscular coordination underlying isokinetic exercise after a stroke and their implications on neurorehabilitation," *J. NeuroEng. Rehabil.*, vol. 18, no. 1, pp. 1–17, Dec. 2021.

[14] W. H. A. Staring, S. Zandvliet, D. de Kam, T. Solis-Escalante, A. C. H. Geurts, and V. Weerdesteyn, "Age-related changes in muscle coordination patterns of stepping responses to recover from loss of balance," *Experim. Gerontology*, vol. 191, Jun. 2024, Art. no. 112424.

[15] C. Dussault-Picard, S. Havashinezhadian, N. A. Turpin, F. Moissenet, K. Turcot, and Y. Cherni, "Age-related modifications of muscle synergies during daily-living tasks: A scoping review," *Clin. Biomechanics*, vol. 113, Mar. 2024, Art. no. 106207.

[16] K. Hoshikawa, T. Yuri, N. Mura, and H. Giambini, "Muscle compensation strategies to maintain glenohumeral joint stability in rotator cuff tears: A cadaveric study," *J. Shoulder Elbow Surg.*, vol. 34, no. 6, p. e414, Jun. 2025.

[17] J. Menze, J. T. Rojas, M. A. Zumstein, S. J. Ferguson, E. De Pieri, and K. Gerber, "Biomechanical evaluation of physical examination tests for rotator cuff tears: A computer simulation study," *J. Shoulder Elbow Surg.*, vol. 34, no. 6, pp. 1574–1582, Jun. 2025, doi: 10.1016/j.jse.2024.09.050.

[18] M. Russo, A. Scano, C. Brambilla, and A. d'Avella, "SynergyAnalyzer: A MATLAB toolbox implementing mixed-matrix factorization to identify kinematic-muscular synergies," *Comput. Methods Programs Biomed.*, vol. 251, Jun. 2024, Art. no. 108217.

[19] P. Ortega-Auriol, W. D. Byblow, A. X. Ren, T. Besier, and A. J. C. McMorland, "The role of muscle synergies and task constraints on upper limb motor impairment after stroke," *Exp. Brain Res.*, vol. 243, no. 1, pp. 1–16, Jan. 2025.

[20] A. Bellitto et al., "Clinical, kinematic and muscle assessment of bilateral coordinated upper-limb movements following cervical spinal cord injury," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 3607–3618, 2023.

[21] D. T. Axford et al., "Three-dimensional continuous muscle moment arm maps for the anatomical shoulder," *J. Biomechanics*, vol. 182, Mar. 2025, Art. no. 112561.

[22] N.-B. Cigercioglu, H. Guney-Deniz, E. Unuvar, F. Colakoglu, and G. Baltaci, "Shoulder range of motion, rotator strength, and upper-extremity functional performance in junior tennis players," *J. Sport Rehabil.*, vol. 30, no. 8, pp. 1129–1137, Nov. 2021.

[23] J. Seol, K. Yoon, and K. G. Kim, "Mathematical analysis and motion capture system utilization method for standardization evaluation of tracking objectivity of 6-DOF arm structure for rehabilitation training exercise therapy robot," *Diagnostics*, vol. 12, no. 12, p. 3179, Dec. 2022.

[24] L. U. Giuseppe et al., "Scapular dyskinesis: From basic science to ultimate treatment," *Int. J. Environ. Res. Public Health*, vol. 17, no. 8, p. 2974, Apr. 2020.

[25] A. M. Cools et al., "Electromyographic analysis of selected shoulder muscles during a series of exercises commonly used in patients with symptomatic degenerative rotator cuff tears," *J. Shoulder Elbow Surg.*, vol. 29, no. 10, pp. e361–e373, Oct. 2020.

[26] N. Zheng and Y. Li, "Recognition of muscular activity in complex noise-contaminated myoelectric signals: An adaptive two-step approach," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–13, 2024.

[27] L. H. Ting and J. L. McKay, "Neuromechanics of muscle synergies for posture and movement," *Current Opinion Neurobiol.*, vol. 17, no. 6, pp. 622–628, Dec. 2007.

[28] C. Huang, X. Chen, S. Cao, B. Qiu, and X. Zhang, "An isometric muscle force estimation framework based on a high-density surface EMG array and an NMF algorithm," *J. Neural Eng.*, vol. 14, no. 4, Aug. 2017, Art. no. 046005.

[29] J. Wei, W. Bai, T. Liu, and X. Tian, "Functional connectivity changes during a working memory task in rat via NMF analysis," *Frontiers Behav. Neurosci.*, vol. 9, Jan. 2015, Art. no. 2.

[30] K. Gui and D. Zhang, "Influence of locomotion speed on biomechanical subtask and muscle synergy," *J. Electromyogr. Kinesiol.*, vol. 30, pp. 209–215, Oct. 2016.

[31] G. Cappellini, Y. P. Ivanenko, R. E. Poppele, and F. Lacquaniti, "Motor patterns in human walking and running," *J. Neurophysiol.*, vol. 95, no. 6, pp. 3426–3437, Jun. 2006.

[32] J. Huegel, A. A. Williams, and L. J. Soslowsky, "Rotator cuff biology and biomechanics: A review of normal and pathological conditions," *Current Rheumatol. Rep.*, vol. 17, no. 1, pp. 1–9, Jan. 2015.

[33] F. Dyrna et al., "Relationship between deltoid and rotator cuff muscles during dynamic shoulder abduction: A biomechanical study of rotator cuff tear progression," *Amer. J. Sports Med.*, vol. 46, no. 8, pp. 1919–1926, Jul. 2018.

[34] E. J. D. Veen et al., "Compensatory movement patterns are based on abnormal activity of the biceps brachii and posterior deltoid muscles in patients with symptomatic rotator cuff tears," *Clin. Orthopaedics Rel. Res.*, vol. 479, no. 2, pp. 378–388, Feb. 2021.

[35] A. Yeung, A. Fernando, M. Patel, L. Gatto, and D. C. Ackland, "Muscle and joint function in the rotator cuff deficient shoulder," *J. Orthopaedic Res.*, vol. 42, no. 10, pp. 2131–2139, Oct. 2024.

[36] E. Croci et al., "Load-induced increase in muscle activity during 30° abduction in patients with rotator cuff tears and control subjects," *J. Orthopaedics Traumatol.*, vol. 24, no. 1, p. 41, Aug. 2023.

[37] F. C. L. de Oliveira, L. J. Bouyer, A. L. Ager, and J.-S. Roy, "Electromyographic analysis of rotator cuff muscles in patients with rotator cuff tendinopathy: A systematic review," *J. Electromyogr. Kinesiol.*, vol. 35, pp. 100–114, Aug. 2017.

[38] B. T. Kelly et al., "Differential patterns of muscle activation in patients with symptomatic and asymptomatic rotator cuff tears," *J. Shoulder Elbow Surg.*, vol. 14, no. 2, pp. 165–171, Mar. 2005.

[39] A. d'Avella, L. Fernandez, A. Portone, and F. Lacquaniti, "Modulation of phasic and tonic muscle synergies with reaching direction and speed," *J. Neurophysiol.*, vol. 100, no. 3, pp. 1433–1454, Sep. 2008.

[40] A. d'Avella, A. Portone, L. Fernandez, and F. Lacquaniti, "Control of fast-reaching movements by muscle synergy combinations," *J. Neurosci.*, vol. 26, no. 30, pp. 7791–7810, Jul. 2006.

[41] J. J. Kutch and F. J. Valero-Cuevas, "Challenges and new approaches to proving the existence of muscle synergies of neural origin," *PLoS Comput. Biol.*, vol. 8, no. 5, May 2012, Art. no. e1002434.

[42] K. M. Steele, M. C. Tresch, and E. J. Perreault, "The number and choice of muscles impact the results of muscle synergy analyses," *Frontiers Comput. Neurosci.*, vol. 7, Nov. 2013, Art. no. 105.

[43] C. L. Banks, M. M. Pai, T. E. McGuirk, B. J. Fregly, and C. Patten, "Methodological choices in muscle synergy analysis impact differentiation of physiological characteristics following stroke," *Frontiers Comput. Neurosci.*, vol. 11, Aug. 2017, Art. no. 78.

[44] B. R. Shuman, M. H. Schwartz, and K. M. Steele, "Electromyography data processing impacts muscle synergies during gait for unimpaired children and children with cerebral palsy," *Frontiers Comput. Neurosci.*, vol. 11, Jun. 2017, Art. no. 50.

[45] K. Kubota et al., "Usefulness of muscle synergy analysis in individuals with knee osteoarthritis during gait," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 239–248, 2021.

[46] Z. Pan, L. Liu, X. Li, and Y. Ma, "Characteristics of muscle synergy and anticipatory synergy adjustments strategy when cutting in different angles," *Gait Posture*, vol. 107, pp. 114–120, Jan. 2024.

[47] M. Hirashima and T. Oya, "How does the brain solve muscle redundancy? Filling the gap between optimization and muscle synergy hypotheses," *Neurosci. Res.*, vol. 104, pp. 80–87, Mar. 2016.

[48] S. Aoi and T. Funato, "Neuromusculoskeletal models based on the muscle synergy hypothesis for the investigation of adaptive motor control in locomotion via sensory-motor coordination," *Neurosci. Res.*, vol. 104, pp. 88–95, Mar. 2016.

[49] L. Favard, D. Katz, M. Colmar, T. Benkalfate, H. Thomazeau, and S. Emily, "Total shoulder arthroplasty—Arthroplasty for glenohumeral arthropathies: Results and complications after a minimum follow-up of 8years according to the type of arthroplasty and etiology," *Orthopaedics Traumatol., Surg. Res.*, vol. 98, no. 4, pp. S41–S47, Jun. 2012.